

- A. Access Switches: The Access switch shall support & include the following items:
1. 10/100/1000BaseT Access Switches with dual 10G uplinks (fiber), No. of ports as per drawings. [Comply](#)
 2. Support for PoE+, IEEE 802.3at on all ports. [Comply](#)
 3. 160 Gbps Switching Fabric. [Comply](#)
 4. FIPS 140-2 certified. [Comply](#)
 5. Support and include OSPF and Multicast.
 6. Ensures data confidentiality with Layer 2 encryption, IEEE 802.1AE on all ports. [Comply](#)
 7. Support for Data Stacking using 64 Gbps. [Comply](#)
 8. Support for Sharing power supplies of the Switches in the stack. [Comply](#)
 9. Full Hardware support for IPv6. [Comply](#)
 10. Operational cost optimization by measuring actual power consumption of the PoE devices, reporting, and reducing energy consumption across the network. [Comply](#)
 11. Support for Smart Multicast to enhance efficiency by putting each video packet onto the backplane only once. [Comply](#)
- B. Distribution Switches: The Distribution switch shall support & include the following items: [Comply but Fixed switch quoted](#)
1. At least 5 Billion Packets/Second.
 2. Chassis should have at least 35 years MTBF.
 3. At least 550 Gbps / Slot.
 4. 10G/40G and 100G Modules.
 5. Layer 3 Unicast Routing, OSPF and BGP.
 6. Layer 3 Multicast Routing, Bidirectional Protocol Independent Multicast.
 7. NEBS Compliant.
 8. MPLS Layer 2 and Layer 3 Support, MPLS TE and RSVP over TE.
 9. Virtualization of up to 8 Virtual Switches on the same chassis.
 10. Each Switch shall have enough IO Modules to support connecting all Access Switches using 2 x 10Gbps ports each and connect to both

Core Switches using 2 x 40 Gbps ports. At least 25% expandability should be accounted for.

C. Core Switches: The Core switch shall support & include the following items:

Comply but Fixed switch quoted

1. At least 5 Billion Packets/Second.
2. Chassis should have at least 35 years MTBF.
3. 10G 1000Base LX fiber ports, No. of ports as per drawings.
4. At least 550 Gbps / Slot.
5. 10G/40G and 100G LR Modules.
6. Layer 3 Unicast Routing, OSPF and BGP.
7. Layer 3 Multicast Routing, Bidirectional Protocol Independent Multicast.
8. NEBS Compliant.
9. Support for Fiber Channel over Ethernet.
10. MPLS Layer 2 and Layer 3 Support, MPLS TE and RSVP over TE.
11. Virtualization of up to 8 Virtual Switches on the same chassis.
12. Tag based access list.
13. Each Switch should have IO Modules to support connecting all Distribution Switches using 2 x 40Gbps ports each and connect to other Core Switch using 2 x 40 Gbps ports. 25 % expandability should be accounted for.

D. Wireless Access Points: The Access points shall support & include the following items: Comply

1. Shall support multiple-input multiple-output (MIMO) technology providing at least six times the throughput of existing 802.11a/g networks,
2. Secure and reliable wireless connections,
3. 802.11n performance with standard 802.3af Power over Ethernet (PoE),
4. 20- and 40-MHz channels,
5. Data rates up to 300 Mbps,
6. At least 128 MB DRAM, 32 MB flash,
7. Compliant to the following standards:

- 802.11i, Wi-Fi Protected Access 2 (WPA2), WPA
- 802.1X
- Advanced Encryption Standards (AES), Temporal Key Integrity Protocol (TKIP)
- Supports the following EAP types:
- Extensible Authentication Protocol-Transport Layer Security (EAP-TLS)
- EAP-Tunneled TLS (TTLS) or Microsoft Challenge Handshake Authentication Protocol Version 2 (MSCHAPv2)
- Protected EAP (PEAP) v0 or EAP-MSCHAPv2
- Extensible Authentication Protocol-Flexible Authentication via Secure Tunneling (EAP-FAST)
- PEAPv1 or EAP-Generic Token Card (GTC)
- EAP-Subscriber Identity Module (SIM)
- The access point should have at least one 10/100/1000BASE-T autosensing for uplink and one console port for management.
- All the manuals, console cable, power cable, mounting kit and any other needed accessories; drivers and software should be supplied with each access point.

Comply with note

E. Network Management: Required is a management platform that integrates all of the above components, namely Routing / Switching/ Firewalls / IPS / Wireless with the following characteristics: **Hardware WLC (Huawei AC6508) quoted for AP management strictly AS PER BOQ. Dedicated full-network NMS software is not in BOQ**

1. Centralized fault and event browser (consolidated, syslog, traps, and events and alarms).
2. Embedded troubleshooting workflow for quick problem isolation and remediation.
3. Simple Network Management Protocol (SNMP)-based polling to identify availability and performance issues.
4. Configuration backup, software image management, compliance, and change management required to maintain and update network devices.
5. Complete and thorough inventory of equipment details - chassis, module, and interface.
6. Provides a single menu for discovery status, device status, user tracking, and inventory dashboards.
7. Reports are centralized in a single menu, simplifying navigation and access to detailed reports and information.
8. Reports can be easily scheduled to run immediately or daily, weekly, or monthly and can be viewed online or optionally exported to a file or comma-separated value (CSV) file.
9. Dashboards provide up-to-date status on configuration, inventory, and monitoring for troubleshooting purposes.